

SoulScope Constellation Dimension Registry v0.1

Status: Canonical scientific implementation companion — final architecture, calibration pending

Governing authority: SoulScope Canon v1.3

Upstream: Evidence Marker Registry v0.1

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1. Purpose

The Dimension Registry defines the 16 functional parameters that form the four SoulScope Constellations.

A Dimension is not an emotion score.

A Dimension is not a personality trait.

A Dimension is a bounded, uncertainty-aware inference from independently qualified Evidence families.

Canonical path:

Evidence → Dimension posterior → Constellation geometry.

2. Dimension classes

Each Dimension is assigned one scientific implementation class.

D1 — DESCRIPTIVE_FUNCTIONAL

The construct is closely tied to observable acoustic organization and may become production-eligible after measurement/reliability validation.

D2 — CONSTRUCT_CALIBRATION_REQUIRED

The construct extends beyond direct acoustic description and SHALL NOT become production-authoritative until SoulScope validates/calibrates it against an appropriate external criterion.

D3 — CURRENT_PROTOCOL_UNOBSERVABLE

The active three-prompt protocol does not contain the observation necessary to resolve the construct.

Its correct default value is:

3. Current 16-point status

ID	Dimension	Class
COG-P1	Organization	D1
COG-P2	Exploration	D2
COG-P3	Focus Continuity	D1
COG-P4	Processing Demand	D2
REG-P1	Activation	D2
REG-P2	Stability	D1
REG-P3	Flexibility	D2
REG-P4	Recovery	D3
CAP-P1	Mobilization	D1
CAP-P2	Reserve	D3
CAP-P3	Effort Cost	D2
CAP-P4	Sustainability	D1
EXP-P1	Range	D1
EXP-P2	Openness	D2
EXP-P3	Restraint	D2
EXP-P4	Relational Availability	D3

Therefore:

$$6 D1 + 7 D2 + 3 D3 = 16.$$

The class does not determine whether a Dimension is interesting.

It determines what evidence is required before SoulScope may responsibly resolve it.

4. Standard Dimension object

Every Dimension SHALL contain:

```
dimension_id
dimension_version

canonical_name
internal_operational_name
scientific_class

prompt_scope

required_evidence_families[]
optional_evidence_families[]

supporting_evidence_ids[]
contradicting_evidence_ids[]
missing_required_evidence[]
active_confounds[]

posterior_mean
posterior_lower
posterior_upper

confidence
evidence_coverage
baseline_trust
contradiction
coherence
momentum

resolution_status
resolution_reason

rule_version
model_version
```

5. General resolution requirements

Unless specifically overridden:

D1

Requires:

- sufficient qualified source material;
- at least two compatible Evidence Markers;
- no invalidating quality condition;
- required Evidence-family coverage.

D2

Requires everything above **plus**:

- active validated construct model;
- approved calibration version;
- required external-validation gate completed.

Without a validated model:

RESEARCH_ESTIMATE

may be stored internally but SHALL NOT masquerade as production-authoritative science.

D3

Returns:

UNRESOLVED

under the current protocol regardless of apparent acoustic pattern.

6. COG-P1 — Organization

Canonical name: Organization

Internal name: VOCAL_TEMPORAL_ORGANIZATION

Class: D1

Definition

Degree to which observable speech-production timing and modulation form a comparatively coherent and sustained pattern during the scan.

Required Evidence families

TIM
DYN

Primary supportive Evidence

EV_TIM_007 SPEECH_RUN_STRUCTURE
EV_TIM_008 OUTPUT_CONTINUITY
EV_TIM_009 TEMPORAL_FRAGMENTATION
EV_DYN_001 WITHIN_PROMPT_DRIFT
EV_DYN_007 MULTI_FAMILY_COHERENCE
EV_DYN_008 MULTI_FAMILY_DISCORDANCE

Higher-expression pattern

Generally supported by:

- stronger observable continuity;
- coherent timing structure;
- lower disruptive fragmentation;
- lower unexplained multi-family discordance.

Lower-expression pattern

Generally supported by:

- greater observable temporal fragmentation;
- shorter/disrupted speech runs;
- pronounced irregularity not explained by quality;
- greater cross-family discordance.

Important boundary

High Organization does not mean:

- organized thoughts;
- intelligence;
- executive functioning;
- mental health;
- good performance.

Low Organization does not mean the opposites.

Example user translation

"Across these responses, your vocal timing stayed comparatively organized and continuous."

or:

"This scan showed more reorganization and interruption in the timing of your responses."

7. COG-P2 — Exploration

Canonical name: Exploration

Internal name: VOCAL_MODULATORY_EXPLORATION

Class: D2

Definition

Breadth and diversity of non-random vocal modulation expressed during open-ended speech and across changing contexts.

Required prompts

P1
plus at least one of P2/P3

Required Evidence families

PRO
TIM
DYN

Primary Evidence

EV_PRO_002 PITCH_RANGE_SHIFT
EV_PRO_003 PITCH_VARIABILITY_SHIFT
EV_ENG_002 LOUDNESS_RANGE_SHIFT
EV_SPE_002 SPECTRAL_DYNAMICS_SHIFT
EV_DYN_003 CONTEXT_RECONFIGURATION
EV_DYN_005 MODULATION_BREADTH
EV_DYN_006 CONTEXTUAL_DEPLOYMENT_SHIFT

High-expression candidate

Broad, non-random modulation deployed across multiple acoustic families and/or contexts.

Low-expression candidate

Comparatively narrow modulation across compatible contexts.

Why D2

Acoustic modulation cannot establish curiosity, cognitive exploration, creativity, or openness of thought.

Before production use, SoulScope must validate whether this construct adds stable explanatory value beyond EXP-P1 Range.

Prohibited meanings

curiosity
creativity
open-mindedness
intelligence
imagination
novelty seeking

8. COG-P3 — Focus Continuity

Canonical name: Focus Continuity

Internal name: SPEECH_OUTPUT_CONTINUITY

Class: D1

Definition

Degree to which ongoing speech production remains continuous rather than becoming materially interrupted or fragmented during the observed task.

Required Evidence

TIM

DYN is strongly preferred.

Primary Evidence

EV_TIM_002 PAUSE_LOAD_SHIFT
EV_TIM_003 PAUSE_DURATION_SHIFT
EV_TIM_004 PAUSE_DENSITY_SHIFT
EV_TIM_007 SPEECH_RUN_STRUCTURE
EV_TIM_008 OUTPUT_CONTINUITY
EV_TIM_009 TEMPORAL_FRAGMENTATION
EV_DYN_009 SHORT_INTERVAL_MAINTENANCE

Prohibited meanings

attention span
ADHD
ability to concentrate
mental focus
executive function
cognitive impairment

Example language

“Your speech flow stayed comparatively continuous through this part of the scan.”

9. COG-P4 — Processing Demand

Canonical name: Processing Demand

Internal name: SPEECH_PRODUCTION_DEMAND

Class: D2

Definition

Convergent acoustic and temporal evidence consistent with increased observable cost or demand in generating and maintaining speech under the administered task.

Minimum independent families

At least:

3

from:

TIM
PHO
PRO
SPE
DYN

Candidate Evidence

EV_TIM_002 PAUSE_LOAD_SHIFT
EV_TIM_005 SPEECH_RATE_SHIFT
EV_TIM_006 ARTICULATION_RATE_SHIFT
EV_TIM_009 TEMPORAL_FRAGMENTATION
EV_PHO_001 CEPSTRAL_SHIFT
EV_PHO_004 PHONATORY_STABILITY
EV_SPE_001 SPECTRAL_BALANCE_SHIFT
EV_DYN_007 MULTI_FAMILY_COHERENCE
EV_DYN_008 MULTI_FAMILY_DISCORDANCE

Critical rule

There is no canonical rule:

$$CPP \downarrow = ProcessingDemand \uparrow$$

or:

$$Pauses \uparrow = ProcessingDemand \uparrow .$$

The parameter requires a calibrated multivariate model.

Why D2

Controlled cognitive-load experiments show that acoustic voice production can vary with manipulated load, including cepstral and spectral measures, but the SoulScope autobiographical prompts are **not controlled cognitive-load manipulations**.

Therefore the user-facing construct remains bounded to speech-production demand during the scan.

Prohibited meanings

brain workload
low intelligence
confusion
cognitive deficit
mental illness
impaired thinking

10. REG-P1 — Activation

Canonical name: Activation

Internal name: ACOUSTIC_MOBILIZATION_STATE

Class: D2

Definition

Degree of coordinated acoustic mobilization relative to compatible reference conditions.

Required Evidence families

At least three independent families from:

PRO
ENG
TIM
PHO
SPE
DYN

Candidate Evidence

EV_PRO_001 PITCH_LEVEL_SHIFT
EV_PRO_002 PITCH_RANGE_SHIFT
EV_ENG_001 LOUDNESS_LEVEL_SHIFT
EV_TIM_005 SPEECH_RATE_SHIFT
EV_PHO_001 CEPSTRAL_SHIFT
EV_SPE_001 SPECTRAL_BALANCE_SHIFT
EV_DYN_007 MULTI_FAMILY_COHERENCE

Critical rule

No permanent direction such as:

$$F0 \uparrow \Rightarrow \textit{Activation} \uparrow$$

shall be canonical without calibration.

Activation must be learned and validated as a multivariate construct.

Prohibited meanings

stress
anxiety
fear
excitement
sympathetic nervous system measurement
arousal certainty

11. REG-P2 — Stability

Canonical name: Stability

Internal name: VOCAL_PATTERN_STABILITY

Class: D1

Definition

Degree of short-timescale consistency versus volatility in qualified vocal organization.

Required Evidence families

At least two of:

PRO
TIM
PHO
DYN

Primary Evidence

EV_PRO_003 PITCH_VARIABILITY_SHIFT
EV_TIM_007 SPEECH_RUN_STRUCTURE
EV_TIM_008 OUTPUT_CONTINUITY
EV_PHO_004 PHONATORY_STABILITY
EV_DYN_002 WITHIN_PROMPT_VOLATILITY
EV_DYN_008 MULTI_FAMILY_DISCORDANCE
EV_DYN_009 SHORT_INTERVAL_MAINTENANCE

Important rule

$$Stability / Flexibility^{-1}$$

A pattern may be both stable within conditions and highly flexible across conditions.

Prohibited meanings

emotional stability
psychological stability
resilience
mental health

12. REG-P3 — Flexibility

Canonical name: Flexibility

Internal name: CONTEXTUAL_VOCAL_RECONFIGURATION

Class: D2

Definition

Degree to which the vocal pattern reorganizes coherently as the administered context changes.

Required prompts

P1 + P2 + P3

Required Evidence

EV_DYN_003 CONTEXT_RECONFIGURATION
EV_DYN_007 MULTI_FAMILY_COHERENCE

plus at least one of:

PRO
ENG
TIM
PHO
SPE

Critical rule

More change does not automatically mean more Flexibility.

A highly volatile, incoherent pattern is not automatically flexible.

Candidate Flexibility should reflect:

context-sensitive change + coherence – unexplained volatility.

Why D2

SoulScope must empirically learn what degree and form of $P1 \rightarrow P2 \rightarrow P3$ reconfiguration represents a reproducible functional construct.

Prohibited meanings

resilience
coping ability
adaptability in life
emotional intelligence
psychological flexibility

13. REG-P4 — Recovery

Canonical name: Recovery

Internal name: POST_CHALLENGE_RETURN

Class: D3

Required observation

A defined challenge followed by an explicit recovery interval.

The active protocol does not contain such a condition.

P3 is a future-oriented prompt.

It is not a recovery phase.

Therefore:

```
status = UNRESOLVED
reason = NO_RECOVERY_COMPATIBLE_CONDITION
```

The similarity:

$$P3 \approx P1$$

is insufficient to claim recovery.

Research on vocal recovery uses explicit loading and subsequent recovery observations, reinforcing the need for a true recovery condition rather than inferring recovery from a different task [S6].

14. CAP-P1 — Mobilization

Canonical name: Mobilization

Internal name: VOCAL_OUTPUT_RECRUITMENT

Class: D1

Definition

Degree of observable recruitment or change in vocal output as task context becomes personally salient.

Required comparison

At least one:

P2 vs P1
P3 vs P1

Candidate Evidence

EV_PRO_002 PITCH_RANGE_SHIFT
EV_ENG_001 LOUDNESS_LEVEL_SHIFT
EV_ENG_002 LOUDNESS_RANGE_SHIFT
EV_TIM_005 SPEECH_RATE_SHIFT
EV_TIM_006 ARTICULATION_RATE_SHIFT
EV_DYN_003 CONTEXT_RECONFIGURATION
EV_DYN_007 MULTI_FAMILY_COHERENCE

Boundary

Mobilization is a description of observable vocal recruitment.

Low Mobilization does not imply low capacity.

Prohibited meanings

motivation
willpower
energy reserves
ambition
engagement certainty

15. CAP-P2 — Reserve

Canonical name: Reserve

Internal name: POST_DEMAND_REMAINING_CAPACITY

Class: D3

Required observation

A validated demand/load structure followed by additional performance demand sufficient to evaluate remaining capacity.

The current protocol does not provide this.

Therefore:

```
status = UNRESOLVED
reason = NO_RESERVE_COMPATIBLE_LOAD_PROTOCOL
```

No combination of normal 30-second autobiographical speech measurements may override this rule.

16. CAP-P3 — Effort Cost

Canonical name: Effort Cost

Internal name: ACOUSTIC_OUTPUT_COST

Class: D2

Definition

Convergent acoustic and temporal disruption or compensation associated with maintaining speech output during the scan.

Candidate Evidence families

At least three from:

```
TIM
PHO
PRO
ENG
SPE
DYN
```

Candidate Evidence

```
EV_TIM_002 PAUSE_LOAD_SHIFT
EV_TIM_007 SPEECH_RUN_STRUCTURE
```


EV_TIM_009 TEMPORAL_FRAGMENTATION
EV_PHO_001 CEPSTRAL_SHIFT
EV_PHO_004 PHONATORY_STABILITY
EV_DYN_001 WITHIN_PROMPT_DRIFT
EV_DYN_009 SHORT_INTERVAL_MAINTENANCE

Critical rule

Acoustics do not directly measure the person's subjective sense of effort.

Until calibrated, the scientific interpretation is:

observable acoustic/timing cost associated with maintaining output.

Prohibited meanings

subjective effort certainty
fatigue diagnosis
burnout
physical energy expenditure
stress level

17. CAP-P4 — Sustainability

Canonical name: Sustainability

Internal name: SHORT_INTERVAL_OUTPUT_MAINTENANCE

Class: D1

Definition

Degree to which observable vocal organization is maintained over the limited time interval actually recorded.

Required Evidence

EV_DYN_001 WITHIN_PROMPT_DRIFT
EV_DYN_009 SHORT_INTERVAL_MAINTENANCE

plus one or more:

TIM
PHO
PRO
ENG

Scope

Maximum claim:

within the observed approximately 30-second response interval(s).

Prohibited meanings

general endurance
chronic energy
burnout
long-term stamina
health capacity

18. EXP-P1 — Range

Canonical name: Range

Internal name: ACOUSTIC_MODULATION_RANGE

Class: D1

Definition

Breadth of qualified acoustic modulation available in the active scan.

Required Evidence families

At least two of:

PRO
ENG
TIM
SPE
DYN

Primary Evidence

EV_PRO_002 PITCH_RANGE_SHIFT
EV_ENG_002 LOUDNESS_RANGE_SHIFT
EV_SPE_002 SPECTRAL_DYNAMICS_SHIFT
EV_DYN_005 MODULATION_BREADTH

Preferred interpretation

Range is one of SoulScope's most directly measurable Constellation dimensions.

Prohibited meanings

emotional depth
expressiveness as personality
social skill
empathy

19. EXP-P2 — Openness

Canonical name: Openness

Internal name: EXPRESSIVE_RANGE_DEPLOYMENT

Class: D2

Definition

Degree to which available acoustic modulation is actually deployed across the observed contexts.

Prerequisite

EXP-P1 Range must itself have sufficient evidence.

Primary Evidence

EV_DYN_003 CONTEXT_RECONFIGURATION
EV_DYN_005 MODULATION_BREADTH
EV_DYN_006 CONTEXTUAL_DEPLOYMENT_SHIFT

plus compatible:

PRO
ENG
TIM

Critical distinction

Range = what modulation is observable

whereas:

Openness = how that available modulation is deployed.

Prohibited meanings

emotional openness
honesty
vulnerability
personality openness
trust
willingness to disclose
receptivity

20. EXP-P3 — Restraint

Canonical name: Restraint

Internal name: CONTEXTUAL_MODULATION_COMPRESSION

Class: D2

Definition

Persistent relative compression of acoustic modulation despite evidence that a broader range is available elsewhere in the scan or compatible personal history.

Critical prerequisite

SoulScope SHALL NOT infer Restraint merely because a person displays narrow acoustic range.

There must be evidence of **available range** against which relative compression can be established.

Conceptually:

Restraint evidence = available range + context-specific compression.

If available range is unknown:

UNRESOLVED

Primary Evidence

EV_PRO_002 PITCH_RANGE_SHIFT
EV_ENG_002 LOUDNESS_RANGE_SHIFT
EV_DYN_005 MODULATION_BREADTH
EV_DYN_006 CONTEXTUAL_DEPLOYMENT_SHIFT

Prohibited meanings

repression
withholding
defensiveness
secrecy
dishonesty
avoidance
social inhibition

21. EXP-P4 — Relational Availability

Canonical name: Relational Availability

Internal name: INTERACTIONAL_VOCAL_RESPONSIVENESS

Class: D3

Required observation

Actual interaction with:

- another speaker; or
- an approved interactive stimulus capable of measuring responsiveness/adaptation.

The active scan is monologic.

Therefore:

```
status = UNRESOLVED
reason = NO_RELATIONAL_OBSERVATION
```

Expressiveness during a monologue is insufficient.

22. Posterior representation

For resolved Dimension d :

$$D_d = \{\mu, L, U, C, E, B, K, H, M, R\}.$$

Where:

Symbol	Meaning
μ	estimated tendency
L, U	uncertainty interval
C	confidence
E	Evidence coverage
B	reference trust
K	contradiction burden
H	coherence
M	temporal/contextual momentum
R	resolution status

23. Provisional inference interface

The production architecture SHALL support:

$$P(D_d \mid E, Q, B, C, P, V)$$

where:

- E = Evidence;
- Q = signal quality;
- B = reference/baseline;

- C = confounds;
- P = protocol;
- V = active versions.

A candidate implementation may use:

- Bayesian models;
- calibrated generalized linear models;
- hierarchical models;
- interpretable machine learning;
- versioned evidence rules.

The Canon does not mandate one mathematical estimator.

It mandates traceability.

24. Abstention rules

A Dimension SHALL resolve as `UNRESOLVED` when any hard requirement fails.

Canonical reasons include:

```
INSUFFICIENT_VALID_SPEECH
MISSING_REQUIRED_EVIDENCE_FAMILY
LOW_SIGNAL_QUALITY
HIGH_EVIDENCE_CONTRADICTION
UNSTABLE_REFERENCE
BASELINE_INCOMPATIBLE
EXTRACTOR_VERSION_INCOMPATIBLE
CONSTRUCT_MODEL_NOT_VALIDATED
NO_RECOVERY_COMPATIBLE_CONDITION
NO_RESERVE_COMPATIBLE_LOAD_PROTOCOL
NO_RELATIONAL_OBSERVATION
ACTIVE_CONFOUND_EXCEEDS_POLICY
```

No unresolved code may be silently converted to the midpoint.

25. Evidence coverage

A Dimension should distinguish:

evidence strength

from:

evidence coverage.

For example, one very large prosodic effect does not compensate for missing required Timing and Phonation families if the active Dimension requires them.

26. Contradiction

Contradiction is first-class information.

Example:

```
Timing evidence -> supports candidate
Phonation evidence -> contradicts candidate
Prosody -> weak/neutral
```

The system SHALL preserve the disagreement.

It may:

- widen the posterior interval;
- reduce confidence;
- create a boundary blend;
- return unresolved.

It SHALL NOT average disagreement into false certainty.

27. First-scan behavior

A person does not need an established personal baseline to receive all possible SoulScope information.

The first scan may use:

- within-prompt dynamics;
- P1 as an open within-session reference;
- P2/P3 contextual contrasts;
- validated compatible population information where explicitly allowed.

However:

missing personal baseline remains missing personal baseline.

Personalized deviation shall become progressively more authoritative only after the baseline protocol is satisfied.

28. Longitudinal behavior

Once an established personal baseline exists:

today → personal historical distribution

becomes an additional Evidence path.

Longitudinal personalization must not silently change the semantic definition of a Dimension.

The same Dimension ID must mean the same construct.

Only the reference quality improves.

29. State-engine input

The State Engine receives the full four-point Constellation object:

```
mean
interval
confidence
coverage
baseline_trust
contradiction
coherence
momentum
resolved_status
```

It does not receive only four scalar values.

A state requiring a Dimension that is unresolved SHALL follow the state registry's missing-evidence rule.

It may not pretend the missing point equals 0.5.

30. Current-protocol consequence

The present three-prompt scan therefore does **not** force four complete four-point shapes.

For example:

```
REG:
  Activation      = candidate/resolved according to validation status
  Stability       = resolved
  Flexibility     = candidate/resolved according to validation status
  Recovery       = unresolved
```

This is scientifically legitimate geometry.

The visualization or narrative must not fill the missing point for aesthetic symmetry.

31. User-language law

Narrative language should identify:

1. observation scope;
2. comparison scope;
3. bounded interpretation;
4. important uncertainty where relevant.

Preferred structure:

“In this scan, your vocal pattern showed ___ compared with ___. That contributed to the ___ result. This describes what appeared in these responses rather than a fixed trait.”

32. Scientific activation gates

Gate A — Measurement verified

Can each upstream parameter be reproduced?

Gate B — Evidence verified

Are Evidence Marker calculations stable and correct?

Gate C — Dimension reliability

Does the Dimension behave reproducibly under compatible repeat conditions?

Gate D — Construct validation

For D2 parameters, does the construct correspond to preregistered external hypotheses?

Gate E — Calibration

Are posterior estimates and confidence well calibrated?

Gate F — Generalization/fairness

Does performance remain acceptable across relevant languages, devices, populations, and speaking conditions?

Only then:

PRODUCTION_ACTIVE

33. Scientific references

[S1] Eyben, F., et al. *The Geneva Minimalistic Acoustic Parameter Set (GeMAPS) for Voice Research and Affective Computing*.

[S2] MacPherson, M. K., et al. *Acoustic Measures of Voice and Physiologic Measures of Autonomic Arousal During Speech as a Function of Cognitive Load*.

[S3] Abur, D., et al. *Acoustic Measures of Voice and Physiologic Measures of Autonomic Arousal During Speech as a Function of Cognitive Load in Older Adults*.

[S4] Delvaux, V., et al. *An Acoustic Study of Natural Emotional Speech Productions*.

[S5] Brockmann-Bauser, M., et al. *Acoustic Perturbation Measures Improve with Increasing Voice Sound Pressure Level...*

[S6] Hunter, E. J., & Titze, I. R. *Quantifying Vocal Fatigue Recovery: Dynamic Vocal Recovery Trajectories After a Vocal Loading Exercise*. Recovery was measured following an explicit loading task across subsequent recovery observations, supporting SoulScope's requirement for a real recovery condition before resolving REG-P4.

34. Final Dimension law

A Dimension may summarize evidence. It may never outrun it.